

Mortality Patterns and Gaps in HIV Care: Insights from Jamaica, 2021–2023

D Perry, CH Weller, R Brown, K Gordon-Johnson, A Douse-Mullings, R Khan-Francis, A Williams, A Robb-Allen, N Skyers, S Beckford-Jarrett

Introduction

In 2015, the World Health Organization recommended that PLHIV should be given Antiretroviral therapy immediately after diagnosis, which is aimed at reducing the mortality rate. Jamaica adopted the test-and-start approach in 2017. Despite these initiatives, PLHIV still experience challenges in treatment that can affect mortality. This study explores the demographic and clinical characteristics of persons diagnosed in Jamaica from 2021 to 2023, and how these factors influence mortality.

Methodology

We analyzed data from PLHIV 15 years and older, who were diagnosed and enrolled at treatment sites in Jamaica between January 1, 2021, and December 31, 2023, to identify how demographic and clinical characteristics influence mortality. Chi-square tests, Kaplan-Meier Survival Analysis, and Cox Regression were used to examine survival patterns and factors associated with mortality during the study period. The data was extracted on July 4, 2025.

Results

A total of 3,056 individuals aged 15 and older, diagnosed from January 1, 2021, to December 31, 2023, were included in the analysis: 1,591 males and 1,465 females. The majority were adults aged 25-59 years, followed by 15 to 24 years, and 60 and over, respectively. Most of the study population was from the Southeast region, while the Northeast accounted for the smallest portion of the population.

WHO HIV staging was used to classify baseline CD4 results; 16.3% were Stage 3 (CD4 200-349), 18.9% were Stage 4 (CD4 <200), and 10.9% had no CD4 results recorded.

During the study period, 6.6% (201) of the study population were recorded as deceased, 590 (19.3%) were no longer in treatment but presumed alive, 137 (4.5%) had migrated, and 2,128 (69.6%) were alive and in treatment.

		N=3,056		Male		Female	
			%	Count	%	Count	%
Age Group at Registration	15-24	589	19.3%	269	16.9%	320	21.8%
	25-59	2172	71.1%	1177	74.0%	995	67.9%
	60+	295	9.7%	145	9.1%	150	10.2%
Region	Southeast	1600	52.4%	845	53.1%	755	51.5%
	Northeast	328	10.7%	162	10.2%	166	11.3%
	Western	689	22.5%	387	24.3%	302	20.6%
	Southern	439	14.4%	197	12.4%	242	16.5%
Year of HIV Confirmation	2021	1055	34.5%	539	33.9%	516	35.2%
	2022	1020	33.4%	551	34.6%	469	32.0%
	2023	981	32.1%	501	31.5%	480	32.8%
HIV Staging (Baseline Stage 1 (>=500) CD4)		1096	35.9%	453	28.5%	643	43.9%
	Stage 2 (350-499)	551	18.0%	307	19.3%	244	16.7%
	Stage 3 (200-349)	499	16.3%	292	18.4%	207	14.1%
	Stage 4 (<200)	578	18.9%	358	22.5%	220	15.0%
	Not Recorded	332	10.9%	181	11.4%	151	10.3%
Suppression at Last Viral Load	Suppressed	2426	79.4%	1252	78.7%	1174	80.1%
	Unsuppressed	305	10.0%	151	9.5%	154	10.5%
	Not Recorded	325	10.6%	188	11.8%	137	9.4%
Outcomes	Death	201	6.6%	128	7.3%	73	5.6%
	LTFU	590	19.3%	342	19.6%	248	19.0%
	Migrated	137	4.5%	92	5.3%	45	3.4%
	Retained in Care	2128	69.6%	1187	67.9%	941	72.0%
Key Populations	CSW	46	1.5%	30	1.9%	16	1.1%
	MSM	217	7.1%	214	13.5%	3	0.2%
	TGW	3	0.1%	2	0.1%	1	0.1%
	Non-Key Pop/Unknown	2790	91.3%	1345	84.5%	1445	98.6%

Table 1. Demographic and Clinical characteristics of people diagnosed HIV positive in Jamaica, 2017-2024. (n=9,135)

No difference in mortality rates was observed by health region. However, mortality was higher in males than females, persons aged 60 years and older, clients with a Stage 4 count (<200), no recorded baseline CD4, and those diagnosed in 2021. In bivariate analysis, variables associated with a higher risk of mortality were age 25-59 years (HR 5.64, 95% CI: 2.64-12.03 vs 15-24), 60 and older (HR 13.34, 95% CI: 6.03-29.51 vs 15-24), Stage 3 baseline CD4 count (HR 1.72, 95% CI: 1.02-2.91 vs Stage 1), Stage 4 baseline CD4 count (HR 3.01, 95% CI: 1.93-4.68 vs Stage 1), and No CD4 count recorded (HR 10.79, 95% CI: 7.16-16.27).

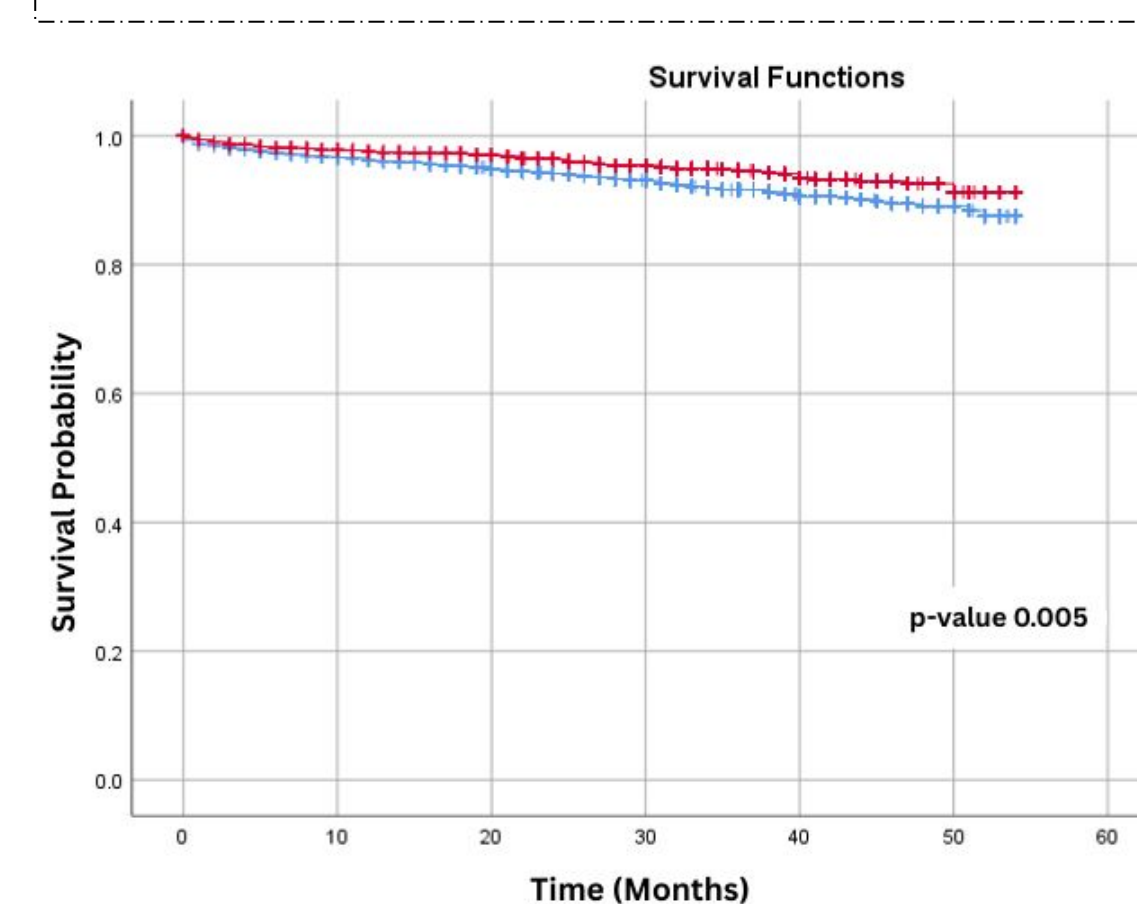


Fig. 1: Kaplan-Meier curves of people diagnosed HIV positive in Jamaica (2021-2023) defined by sex. (n=3,056)

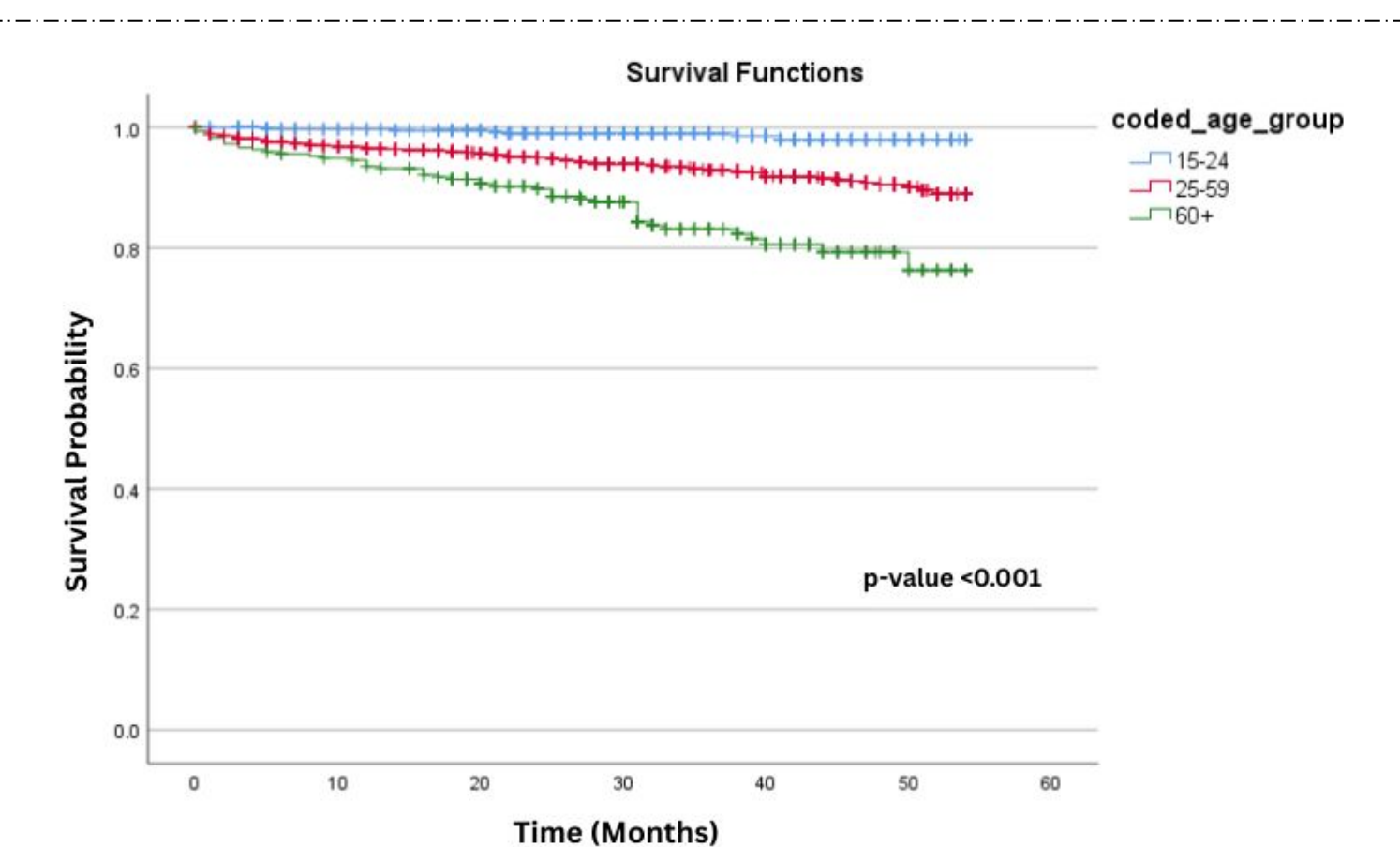


Fig. 2: Kaplan-Meier curves of people diagnosed HIV positive in Jamaica (2021-2023) defined by age group. (n=3,056)

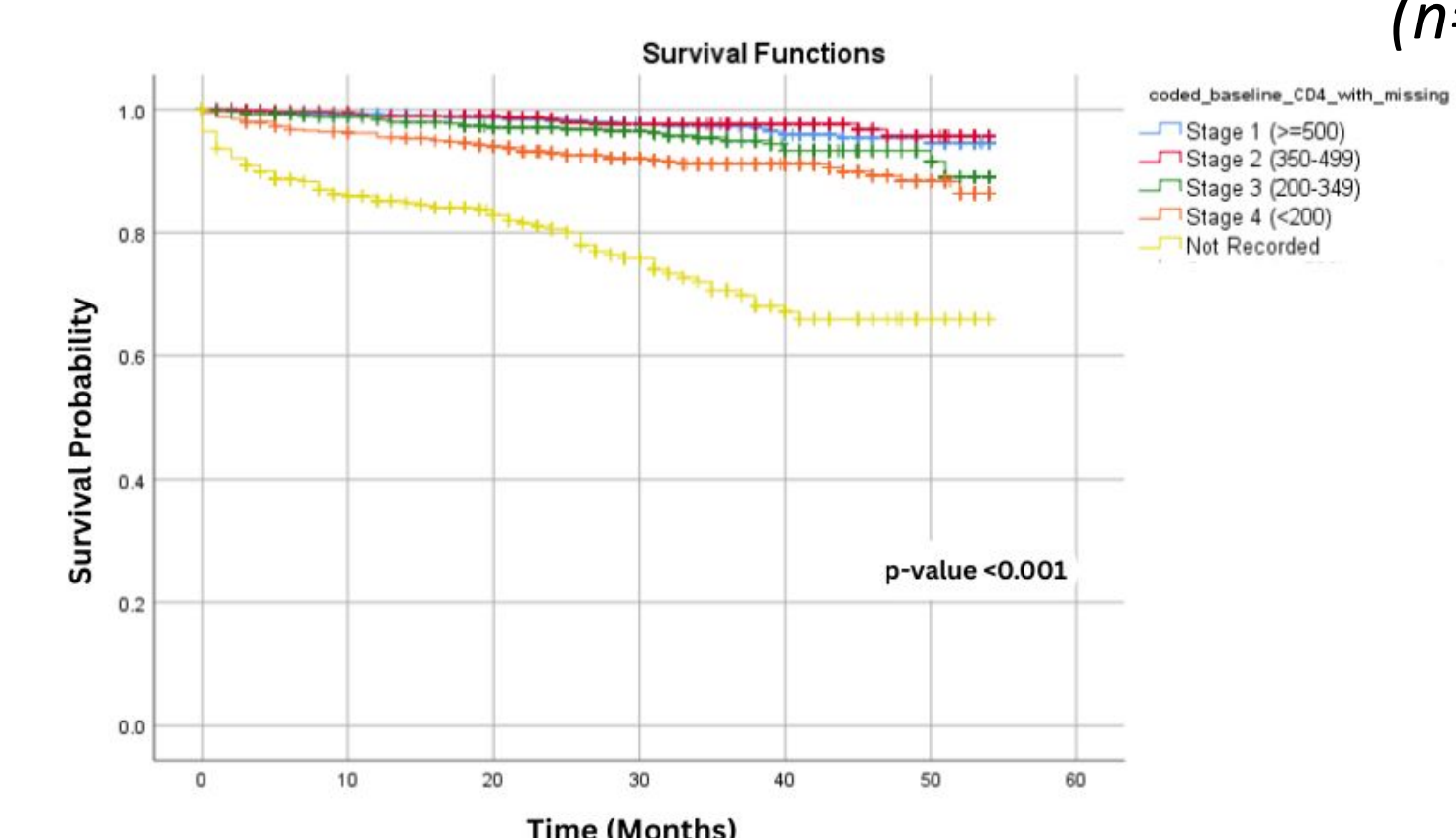


Fig. 3: Kaplan-Meier curves of people diagnosed HIV positive in Jamaica (2021-2023) defined by CD4 Staging at baseline CD4. (n=3,056)

In multivariate analysis, 25-59 years at diagnosis (AHR 4.72, 95% CI: 2.01-10.11), 60 and over at diagnosis (AHR 10.05, 95% CI: 4.53-22.31), Stage 4 baseline CD4 count (AHR 2.38, 95% CI: 1.52-3.72), and No CD4 count recorded (AHR 8.81, 95% CI: 5.82-13.33) remained significantly associated with a higher risk of mortality.

Discussion

Findings from this cohort highlight opportunities to strengthen the national HIV response. Although mortality outcomes have improved compared to earlier cohorts, rates may be underestimated, as 19% were lost to follow-up. Mortality was highest among persons diagnosed at age 60+, clients with advanced immunosuppression (CD4 <200; Stage 4), and clients without a recorded baseline CD4. The association between missing CD4 and poor survival may reflect programmatic gaps in testing and documentation as well as clinical or retention challenges. Program priorities should include targeted HIV testing, ensuring prompt ART initiation, improving retention in care, and strengthening CD4 testing and documentation.

Daniel Perry – Parish TCSO (St. Catherine Health Department) - danielmp@serha.gov.jm

Hannalori Weller – Regional TCSO (Southeast Regional Health Authority) - chweller79@gmail.com

Rose-Ann Brown – Parish TCSO (Kingston & St. Andrew Health Department) - roseann.brown.tcs@gmail.com